

Monochromatic radiation

In physics, **monochromatic radiation** is radiation with a single constant frequency or wavelength.^[1] For electromagnetic radiation, when that frequency is part of the visible spectrum (or near it) the term **monochromatic light** is often used. Monochromatic light is perceived by the human eye as a spectral color.

When monochromatic radiation propagates through vacuum or a homogeneous transparent medium, it remains with a single constant frequency or wavelength; otherwise, it suffers refraction.

Practical monochromaticity

No radiation can be totally monochromatic,^[1] since that would require a wave of infinite duration as a consequence of the Fourier transform's localization property (cf. spectral coherence). In practice, "monochromatic" radiation — even from lasers or spectral lines — always consists of components with a range of frequencies of non-zero width.

Generation

Monochromatic radiation can be produced by a number of methods. Isaac Newton observed that a beam of light from the sun could be spread out by refraction into a fan of light with varying colors; and that if a beam of any particular color was isolated from that fan, it behaved as "pure" light that could not be decomposed further.

When atoms of a chemical element in gaseous state are subjected to an electric current, to suitable radiation, or to high enough temperature, they emit a light spectrum with a set of discrete spectral lines (monochromatic components), that are characteristic of the element. This phenomenon is the basis of the science of spectroscopy, and is exploited in fluorescent lamps and the so-called neon signs.

A laser is a device that generates monochromatic and coherent radiation through a process of stimulated emission.

Nearly monochromatic gravitational waves can be generated by spinning non-spherical neutron stars^[2] or binaries a long time before merger.



Red Green Blue lasers



Properties and uses

When monochromatic radiation is made to interfere with itself, the result can be visible and stable interference fringes that can be used to measure very small distances, or large distances with very high accuracy. The current definition of the metre is based on this technique.^{[3][4]}

In the technique of spectroscopic analysis, a material sample is exposed to monochromatic radiation, and the amount that is absorbed is measured. The graph of absorption as a function of the radiation's frequency is often characteristic of the material's composition. This technique can use radiation ranging from the microwaves, as in rotational spectroscopy, to gamma rays, as in Mössbauer spectroscopy.

See also

- Wave
- Acoustics
- Optics
- Monochromator
- Interferometer
- Diffraction grating
- Dichroic filter
- Monochromatic plane wave
- Newton rings

References

1. Entry "monochromatic light" (<https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100206892>) in the *Oxford Reference* online dictionary. Accessed on 2021-11-22 at
 2. Wette, Karl (2023-11-01). "Searches for continuous gravitational waves from neutron stars: A twenty-year retrospective" (<https://www.sciencedirect.com/science/article/pii/S092765052300066X>). *Astroparticle Physics*. **153** 102880. arXiv:2305.07106 (<https://arxiv.org/abs/2305.07106>). doi:10.1016/j.astropartphys.2023.102880 (<https://doi.org/10.1016%2Fj.astropartphys.2023.102880>). ISSN 0927-6505 (<https://search.worldcat.org/issn/0927-6505>).
 3. Bureau international des poids et mesures (2019): Le système international d'unités (<https://www.bipm.org/documents/20126/41483022/SI-Brochure-9.pdf/fcf090b2-04e6-88cc-1149-c3e029ad8232>), complete brochure, 9th edition.
 4. NIST (2019): *Special Publication 330: The International System of Units (SI) 2019 Edition* (<https://www.nist.gov/pml/special-publication-330>).
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